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# A Validated Black-Box Instrument for Language-Model Dynamics: Reproducing Known Metrics with a Token-Lattice Cellular Automaton

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## Abstract

1 We turn a language model into a *cellular automaton over token space* — a ring  
2 of token cells resampled in place by the model’s own windowed conditional  
3  $p_r(x_i \mid x_{i\pm r})$ , probed by common-random-number damage spreading — and  
4 develop it into a *validated* measurement instrument, on the principle of **valida-**  
5 **tion by reproduction**. On the Domany–Kinzel automaton, where the damage  
6 field is provably the automaton itself, an independent prediction matches ours  
7 **bit-exactly**: zero mismatching cells over  $4096 \times 1500$  updates, with a nonzero  
8 off-line control, calibrating the apparatus with no error bar attached. It also sep-  
9 arates elementary CA rules by whether damage survives ( $d=3.0$ ) and recovers  
10 known Markov matrices. Across Pythia-410m checkpoints it detects a **develop-**  
11 **mental transition**: the token-space Lyapunov exponent goes from seeds disagree-  
12 ing about its *sign* to **every one of 48 post-transition runs being positive**, at two  
13 lattice sizes, all four members of a pre-registered family surviving Benjamini–  
14 Hochberg correction. We then *delimit* it, the result we consider most useful: real  
15 autoregressive generation absorbs *no* injected token error ( $P_{\text{persist}}=1.000$  on three  
16 models), because free generation never resamples a token. These quantities char-  
17 acterise an iterated-resampling construction, not deployed decoding — a boundary  
18 we draw rather than leave to the reader.

## 19 1 Introduction

20 Language models are evaluated almost entirely as functions: given a context, how good is the next-  
21 token distribution? But a masked or autoregressive conditional also *defines a dynamics* — iterate it  
22 in place over a lattice of tokens and one obtains a stochastic cellular automaton whose rule is the  
23 model. The phenomena a CA vocabulary names — damage spreading [Bagnoli et al., 1992], light  
24 cones [Lieb and Robinson, 1972], Lyapunov exponents, attractor censuses [Hanson and Crutchfield,  
25 1997] — are decades old. What has been missing is not a phenomenon but a *measurement instrument*  
26 *whose readings can be trusted*.

27 Our organizing principle is therefore **validation by reproduction** (§3): before turning the appara-  
28 tus on a language model we require it to recover values that are independently known. Damage-  
29 spreading implementations are conventionally validated against fitted critical exponents, which a  
30 wrong implementation can match for the wrong reasons; our strongest rung is instead *exact*, and a  
31 bit-exact identity cannot be passed by a plausible-looking estimator. Only then do we report what  
32 the instrument measures on trained LMs (§4) and, equally, where its readings provably stop applying  
33 (§5, §6).

## 34 2 The instrument

35 **Rule.** A window of  $w=2r+1$  tokens with the centre masked is fed to a masked LM; the tempered  
 36 centre distribution is the kernel  $p_r(x_i \mid x_{i\pm r})$ . For an autoregressive model the window is the  $r$   
 37 cells to the left. **Automaton.**  $N$  ring cells, async random-order single-site Glauber; special tokens  
 38 forbidden as emissions. **CRN coupling.** Twin lattices sharing initial state, update order and uniform  
 39 stream differ only in the factor under test, so the *null* arm must diverge by exactly zero — asserted on  
 40 every backend, because every damage number depends on it. **Confounds.** A degenerate low-entropy  
 41 lattice snaps a perturbation back trivially, so damage  $D$  is normalised by the model’s own diversity  
 42 floor  $D_0$  (twins under *independent* noise, no flip), giving  $D_{\text{norm}}=D/D_0$ . The two necessarily use  
 43 different couplings — a floor sharing the numerator’s is identically zero, since identical twins stay  
 44 identical — but the cost is bounded: sweeping the shared-draw fraction from 0 to 0.9 moves  $D_{\text{norm}}$   
 45 by 4%. A free control disciplines our language: the temperature “phase transition” is a finite-size  
 46 *crossover* ( $\chi_{\text{peak}} \propto 1/N$ ).

## 47 3 Validation: reproducing known metrics

48 We check the apparatus against systems where the answer is known, ordered by how much of the  
 49 instrument’s regime each rung shares (Fig. 1). (1–2) are smooth-limit unit tests and are labelled  
 50 as such: a renormalized tangent estimator recovers the logistic map’s analytic exponent to machine  
 51 precision and a coupled-map lattice matches an exact Benettin reference to  $1.1 \times 10^{-3}$ , but neither is  
 52 evidence about damage spreading, since the twin is re-anchored each step and **a token flip is  $O(1)$** .

53 **(3) Elementary CA: a coarse split.** The identical protocol on 19 rules of *known* class (12 seeds,  
 54 rule as the unit) separates damage that *survives* from damage that *dies*. The discriminating quantity  
 55 is the **ignition probability** — also the directed-percolation order parameter [Grassberger, 1995] —  
 56 not  $\lambda$ , which averages a bimodal mixture. Ordered rules sit at 0.046 (CI [0.000, 0.102]) against  
 57 0.668 (edge) and 0.682 (chaotic): **ordered vs. rest**  $p=0.0$ ,  $d=3.03$ . The three-class ordering does  
 58 *not* survive ( $p=0.470$ ), so we claim only the coarse split, and report it rather than the group-mean  
 59  $\lambda$  ordering we first computed: five of seven ordered rules return exactly  $-0.4 \ln 10$ , the constant a  
 60 log-linear fit yields when damage dies immediately, so that end was never a measurement.

61 **(4) Domany–Kinzel: an exact check.** DK [Domany and Kinzel, 1984] is the only rung that is  
 62 stochastic *and* discrete — the instrument’s own regime — and on its  $p_2=0$  line the damage field be-  
 63 tween two commonly-driven replicas is provably *itself* a DK automaton at the same  $p_1$  [Kohring and  
 64 Schreckenberg, 1992]. We therefore predict the damage trajectory with an independent simulator  
 65 and demand *bit-identity*: across  $p_1 \in \{0.2, \dots, 1.0\}$  on a ring of 4096 over 1500 steps, **zero mis-**  
 66 **matching cells**, against 16 in an off-line control at (0.6, 0.5) where the mapping must break. It runs  
 67 through the same loop as every language-model number below, so indexing, uniform consumption,  
 68 sampling and schedule are verified with no error bar attached; Rule 90 rides inside the same run as  
 69 a second closed form (0.03125 live cells at  $t=1500$ , as obtained). Critical points are a weaker  $\sim 1\%$   
 70 check:  $p_c=0.7065$  on the site-DP line (published 0.705489(4)) and 0.8092 on  $p_2=0$ , the damage  
 71 field independently at 0.8089. The published  $p_2=0$  value is disputed (0.801(2) [Zebende and Penna,  
 72 1994] vs. 0.8087(5) [Hinrichsen et al., 1997]), so we report both and claim neither.

73 **(5) Synthetic Markov sources.** The attractor census recovers each model’s own sparse transition  
 74 matrix (row total-variation 0.22) and not the others (cross 0.95; baseline 0.91).

## 75 4 A developmental transition in token-space criticality

76 Across Pythia-410m checkpoints (step256–step143000, 8 seeds,  $N=48$  and 96; Fig. 2) the cali-  
 77 brated instrument finds token-space dynamics crossing from sub- to super-critical between steps 256  
 78 and 512 at both lattice sizes, and staying positive thereafter.

79 **The effect occupies a temperature window, and the window has a mechanism.** Repeating the  
 80 two ends of the curve at  $T \in \{0.3, 0.5, 0.9, 1.1\}$  (8 seeds each, BH-FDR over the four) recovers  
 81 the transition at  $T=0.5$  ( $p_{\text{BH}}=6 \times 10^{-4}$ ) alongside the  $T=0.7$  of the main grid, and at none of 0.3,  
 82 0.9, 1.1 ( $p_{\text{BH}}=0.59, 0.72, 0.59$ ). The ignition fraction makes that a floor and a ceiling rather than  
 83 a failure: at  $T=0.3$  damage barely propagates at either end ( $0.20 \rightarrow 0.21$ ) and  $\lambda_{\text{ca}}$  hugs zero,  
 84 whereas at  $T=0.9$  and 1.1 the lattice is *already* super-critical before the training being measured

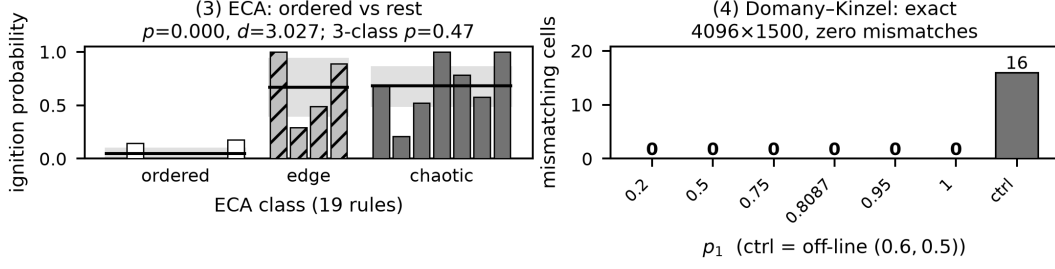


Figure 1: The two rungs that carry the calibration, numbered as in the text. (4) is the load bearing one: the DK damage field is predicted by an independent simulator and matches cell-for-cell, the off-line control beside it showing the test is not vacuous. Rungs (1–2) and (5) are reported in the text.

( $\lambda_{ca} + 0.19$  and  $+0.30$  pre, ignition 0.65 and 0.98), leaving nothing to move; inside the window it swings  $0.23 \rightarrow 0.81$  at  $T=0.5$ . Independent work puts Pythia *generation*’s critical temperature at  $T_c \approx 1$  [Nakaishi et al., 2024]; a different sampler, so a consistent reference point rather than the same measurement — but a ceiling between 0.7 and 0.9 is where it predicts one.

The test was pre-registered (post- vs pre-transition, run-level Mann–Whitney, family =  $\{\lambda_{ca}, D_{norm}\} \times \{N=48, 96\}$ , BH-FDR) with a demotion rule: failure at either size demotes the headline. **All four survive** ( $p_{BH} \leq 2 \times 10^{-5}$ ). From the pre set (256/512) to the plateau (2000/8000/143000),  $\lambda_{ca}$  moves  $+0.0247 \rightarrow +0.1683$  at  $N=48$  ( $d=1.59$ ) and  $+0.0320 \rightarrow +0.1686$  at  $N=96$  ( $d=1.71$ ,  $n_{pre}=15$ : one run is excluded because its damage never ignited, so  $\lambda_{ca}$  is undefined for it).

**The claim is better stated without any pre/post split**, since every effect size depends on where that line is drawn. Before the transition the seeds do not agree on the *sign* of  $\lambda_{ca}$  (6/16 negative at  $N=48$ , 7/15 at  $N=96$ , spanning  $-0.216$  to  $+0.320$ ); after it, **not one of 48 plateau runs is negative**, the minimum being  $+0.107$ . That is ordinal, uses every run, and survived a later change to unignited-run handling that moved every mean.

**The curve is not a step, and both ends can be inflated.** It overshoots at step 1000 by 1.4–22.4%; taking the peak as the post value, or the lowest checkpoint alone as the pre value, inflates the effect ( $1.7\times$  on  $\lambda_{ca}$  for the latter), so both ends use the pre-registered sets. The overshoot survives correction in one of four cells, and that cell is  $D_{norm}$  — on  $\lambda_{ca}$ ,  $+1.4\%$  at  $N=96$  ( $p_{BH}=0.78$ ) — so step 1000 is no distinct phase.

**It is not specific to the model we picked.** Repeating the grid on four Pythia sizes (70m/160m/410m/1b  $\times 6$  checkpoints  $\times 8$  seeds, 192 runs) reproduces the transition in **all four** ( $p_{BH} = 0.015, 0.003, 2 \times 10^{-6}, 1.5 \times 10^{-5}$ ). Its *timing* moves later with size and then saturates: 70m is already super-critical at the earliest checkpoint probed, 160m crosses between steps 128 and 256, and 410m and 1b both cross between 256 and 512. Learning rate is confounded with size across the suite ( $1.0 \times 10^{-3}, 6.0 \times 10^{-4}, 3.0 \times 10^{-4}, 3.0 \times 10^{-4}$  [Biderman et al., 2023]) and the two sizes sharing a rate are the two sharing a bracket, so we report the shift as an ordering, not a size effect. An orthogonal instrument agrees on the onset: Nakaishi et al. [2024] place the emergence of critical structure in Pythia-160m at  $k_c \approx 10^2$  steps by power spectra and part-of-speech correlation — no damage spreading — adjacent to our 128–256 bracket. Both grids are coarse there, so this is agreement to within about a factor of two, not a reproduction. The plateau *level* does not order with size ( $0.162/0.164/0.174/0.166$ ); at four sizes the smallest attainable permutation  $p$  is  $1/4! = 0.042$ , so no capacity axis could have been established, and none is visible. We report that null as the closest thing here to a previously retracted capacity claim.

**It is not a restatement of the loss curve.** The transition sits where everything else in training changes, so  $\lambda_{ca}$  might be an expensive perplexity proxy. It is not. Held-out loss at 26 (model, checkpoint) pairs — the scale grid’s 24 plus 410m at steps 8000 and 143000 — falls *monotonically* at all four sizes while  $\lambda_{ca}$  overshoots at three, and a non-monotone function of a monotone variable is not a monotone transform of it. They also disagree about *where*: the steepest loss descent is the 512–1000 bracket for *every* size, whereas the  $\lambda_{ca}$  crossing moves with size and precedes it. Rank correlation is mostly not significant at six checkpoints ( $\rho$  from  $-0.77$  to  $-0.71$ ), so shape and

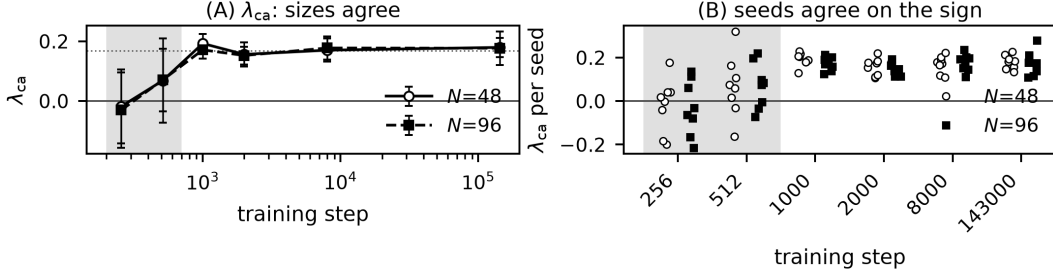


Figure 2: The developmental transition; shading marks the pre-registered pre set. (A)  $\lambda_{ca}$  rises and settles positive, the two lattice sizes on top of each other (levels agree within  $\pm 14\%$ ). (B) Per-seed values: sign disagreement before, consensus after.  $D_{norm}$  has the same shape at a size-dependent level.

location carry this and not the correlation. We use the curve’s *shape* and not its level, the evaluation text not being Pythia’s training data.

**Which quantity carries the claim.** A third lattice size settles it. Over  $N=48/96/192$  the  $\lambda_{ca}$  plateau is 0.168/0.169/0.160 — slope  $N^{-0.04}$ , **intensive across a  $4\times$  range**, varying by 5% of its mean — while  $D_{norm}$  is 0.569/0.306/0.139, slope  $N^{-1.02}$ , i.e.  $1/N$ . That is what the construction predicts:  $\lambda_{ca}$  is a cone-growth *rate* fitted before saturation;  $D_{norm}$  is a density ratio whose numerator stays in a cone while its denominator is delocalised. So  $\lambda_{ca}$  carries the claim and  $D_{norm}$  corroborates at a stated lattice size, never as a lattice-free property. Seed-to-seed variance also collapses  $\sim 3\times$  across the transition (Levene  $p \leq 1.7 \times 10^{-4}$  at both sizes), reported as an observation, not pre-registered; it is the checkpoint means that agree to a few percent, the plateau per-seed CV being 22–25%.  $\lambda_{ca}$  is continuous and unthresholded, so this sharpening is not an artefact of a discontinuous metric [Schaeffer et al., 2023].

## 5 What the damping length measures: free generation absorbs nothing

We tested the instrument’s central assumption — that a ring number says something about the model — on *real autoregressive generation*: two CRN twins from a prompt sharing one uniform stream, one forced to a different token at one position, both continuing freely. The null arm diverges by exactly zero, as asserted.

The result is unambiguous and negative. On pythia-70m, -160m and -410m (32 trials each) the injected error **persists in every single trial** ( $P_{persist}=1.000$ ). Token identity is harsh once an injection shifts alignment, so we also measured distributionally: twin next-token TV, normalised by that between *independent* continuations, settles at  $TV_{norm} \approx 0.97$ .

The mechanism is structural. **Free generation never resamples a token**: once emitted an error stays in context permanently, whereas the ring CA revisits every site, and in-place resampling is what makes healing possible. The damping length,  $D_{norm}$  and the light cone therefore characterise *the iterated-resampling construction* and not decoding — which also explains a light-cone velocity depending on  $r$  but not the model, and the failure in §6. It does *not* touch §4: the construction is held *fixed* across checkpoints — same ring, radius, temperature, coupling, seeds — so a change measured across checkpoints cannot come from the apparatus. The licence is therefore *comparative*, checkpoint against checkpoint, and never absolute: we do not claim any model *is* critical, since that would need the ring, the radius and  $D_0$  to be principled rather than merely fixed, and they are choices.

**Coupling, stated precisely.** Damage spreading is a property of (*model, coupling*), not of the model alone [Kohring and Schreckenberg, 1992, Grassberger, 1995]. Our inverse-CDF sampling is the *monotone* coupling: on DK’s binary alphabet it coincides with the maximal-correlation member of the admissible family [Hinrichsen et al., 1997], which is why the identity above is exact; on a vocabulary it does not, so our LM damage numbers sit inside the family rather than at an extremum (excess disagreement 1.3–5.4%). The choice is justified by replica-independence — each replica’s next state depends on its own state and the shared noise alone — which a pairwise maximal coupling lacks. Relative comparisons are unaffected: the coupling is a common mode. **One fact explains**

165 **both.** Damage-spreading theory was developed on *binary* alphabets — which is why the DK rung is  
 166 exact, and why its coupling result does not survive  $|V| \approx 5 \times 10^4$ . The strongest calibration and the  
 167 sharpest limitation share one cause, so an imported guarantee must be checked against the alphabet.

## 168 6 The instrument’s boundary: token- vs. activation-space

169 A black-box instrument earns its keep if its readings track what internals would otherwise be needed  
 170 to see. Across two model families and two within-model axes they do not: the pairing is family-  
 171 dependent (Pythia  $r=+0.71$ , GPT-2  $r=-0.43$ ; the pooled  $p=0.025$  is a Simpson’s-paradox artifact),  
 172 the temperature axis is confounded, and the radius axis is null because  $\lambda_{ca}(r)$  is model-invariant.  
 173 The reason is structural: the white-box exponent is architectural and *tangent*, while  $\lambda_{ca}$  is *finite*. A  
 174 clean **negative** — the two levels are distinct and non-proxying.

## 175 7 Related work

176 Iterated-LLM views exist qualitatively [Utimula, 2026, Perez et al., 2024, Wang et al., 2025] and  
 177 rigorously for masked-LM Glauber dynamics [Sana et al., 2026]; our novelty is the measurement  
 178 layer, not the framing. Edge-of-chaos measurement is canonical [Langton, 1990, Bertschinger and  
 179 Natschlager, 2004] and already applied to transformers [Tomihari and Karakida, 2025] and LLMs  
 180 [Zhang et al., 2025, Li et al., 2025]; we claim neither the concept nor the criticality $\leftrightarrow$ capability link,  
 181 and absorbing-state scaling laws in deep networks are established [Tamai et al., 2025] — we report  
 182 no universality class. Temperature-driven order–disorder transitions have several prior treatments  
 183 [Ruan et al., 2026, Sun and Haghighat, 2025, George et al., 2025, Nakaishi et al., 2024, Arnold  
 184 et al., 2024]; we claim a calibrated route, not the quantity. Perturbation propagation in *activation*  
 185 space [Petrova and Vejsiu, 2026, Olivieri and Perez Rodrıguez, 2026, Fernando and Guitchounts,  
 186 2026] and over compound-AI graphs [Nilayam and Nayak, 2026] is adjacent but in a different space;  
 187 we avoid “repair” [McGrath et al., 2023, Rushing and Nanda, 2024] and “self-correction” [Petrova  
 188 and Vejsiu, 2026], categorically different from a spatial damping length. Tang et al. [2026] is the  
 189 calibration kin; the census descends from Hanson and Crutchfield [1997].

## 190 8 Limitations

191 Neither construction samples the model’s actual distribution — masked conditionals are globally  
 192 inconsistent, the autoregressive rule is resampled in place on a ring — so all claims are properties  
 193 of the dynamics. Rings are small ( $N \leq 384$ ); the developmental result rests on one model *family*  
 194 (four sizes and three lattice sizes within it, so a second family is the next requirement) and on a  
 195 narrow temperature window, outside which the probe saturates either way (§4).  $D_{\text{norm}}$ ’s absolute  
 196 scale is doubly qualified — it moves with  $N$ , and its numerator’s coupling is not extremal — so only  
 197 its relative behaviour is quoted, and census ground truth is synthetic only. **Statistical scope.** The  
 198 developmental family is pre-registered with BH-FDR at the run level and the ECA test uses the rule  
 199 as its unit;  $\lambda_{ca}$  statistics exclude runs whose damage never ignited, for which it is undefined, with  $n$   
 200 stated. The ladder rests on no hypothesis test and the cross-level negative on sign-consistency rather  
 201 than a surviving  $p$ -value. A previously reported capacity $\rightarrow$ sensitivity axis was pseudoreplicated and  
 202 is retracted.

## 203 9 Responsible use and conclusion

204 This instrument measures a construction built *from* a language model, from sampled outputs alone.  
 205 That is its affordance: it characterises models behind an API, where weights are unavailable and  
 206 activation-space methods cannot reach. It creates no new generative capability. The misuse we con-  
 207 sider most likely is a misreading of our own numbers, which is why §5 is in the body: a damping  
 208 length measured here describes an in-place-resampling ring, and real generation absorbs *no* injected  
 209 token error. These quantities must **not** be cited as evidence about the robustness, error tolerance or  
 210 self-correction of a deployed decoder — a  $\lambda_{ca}$  below zero is not a safety property. That boundary  
 211 generalises: a metric should be made to reproduce known answers before it is believed about un-  
 212 known ones. We offer the ladder as a method, not only as our step one — it demoted our own ECA  
 213 ordering and caught a  $\lambda_{ca}$  reported where it is undefined.

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## 283 A Reproducibility

284 **Models.** bert-tiny/mini/base ( $L=2-24$ ); Pythia-14M-1B [Biderman et al., 2023] and GPT-2  
285 small-xl; Pythia-410m checkpoints step256–step143000. A tiny transformer trained from  
286 scratch on tinyshakespeare provides the corpus-checked census. **Grids.**  $r \in \{1, 2, 4, 8, 16\}$ ,  
287  $T \in \{0.3, \dots, 1.5\}$ ,  $N \in \{48, 96, 192, 384\}$ , 8 seeds for the developmental family. **Definitions.**  
288  $D_{\text{norm}} = D/D_0$  with  $D_0$  the independent-noise no-flip floor;  $\lambda_{\text{ca}}$  the log-slope of damaged-site  
289 count over the pre-saturation window, with the fit window’s constants exposed as parameters and a  
290 named sentinel for never-ignited runs. **Tests.** The exact-zero CRN null is asserted over every back-  
291 end and both update modes through one shared simulation loop; a golden-file regression asserts the  
292 core stays bit-identical; the DK identity is asserted cell-for-cell with an off-line control that must  
293 fail. All code, per-run result files, and figure scripts accompany the submission; every number in the  
294 paper is traceable to a result file.

## NeurIPS Paper Checklist

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Question: Do the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract states the contribution as a *validated* black-box instrument and names the three things it does: reproduce known metrics before measuring LMs (the load-bearing rung being Domany–Kinzel, where agreement is *bit-exact* rather than fitted), detect a developmental transition across Pythia checkpoints, and *delimit* itself — real generation absorbs no injected token error, so the quantities describe an iterated-resampling construction rather than decoding. The smooth-limit rungs (logistic map, coupled-map lattice) are labelled in the abstract and body as arithmetic unit tests, not validation. The criticality phenomena are framed as decades-old (measured, not discovered) and cited to prior work. The cross-level proxy hypothesis is reported as a clean *negative*. No capacity scaling law is claimed — that earlier two-seed gap is retracted and explicitly not built upon.

### 2. Limitations

Question: Does the paper discuss the limitations of the work?

Answer: [Yes]

Justification: The Limitations section states: neither construction samples the model’s actual distribution (masked conditionals are globally inconsistent; the autoregressive rule is consistent but resampled in place on a ring), so all claims are properties of the dynamics; small rings ( $N \leq 384$ ) and one model family for the developmental result, with a second family named as the obvious next requirement;  $D_{\text{norm}}$ ’s absolute scale doubly qualified (it moves with  $N$ , and its numerator’s coupling is not extremal), so only its relative behaviour is quoted; synthetic-only census ground truth; the windowing scheme as a first-class apparatus factor; temperature scales not comparable across constructions. The statistical scope is stated separately, including that  $\lambda_{\text{ca}}$  statistics exclude runs whose damage never ignited — for which the quantity is undefined — with  $n$  stated in each case.

### 3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper contains no formal theorems; it is empirical/measurement work. Connections to the Lieb–Robinson bound and the Bagnoli–Rechtman–Ruffo Lyapunov–velocity relation are cited, not re-derived.

### 4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: All code, model checkpoints (toy), per-run result JSON/NPZ, and figures are supplied as anonymised supplementary material; the Reproducibility appendix gives models, grids, seeds, batch/ring sizes, the diversity-floor and CRN definitions, and the statistical tests. A harness regression test asserts the CRN null coupling is exactly zero.

### 5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions?

Answer: [Yes]

Justification: The anonymised supplement contains the full codebase with a README giving per-phase reproduction commands and measured M1 runtime tables, pinned dependency versions, and idempotent resumable scripts. Every number in the paper is traceable to a committed result file; the analysis scripts assert their emitted group sizes against their declared pre-registered design. A public repository URL will be added at camera-ready if the paper is accepted.

### 6. Experimental setting/details

Question: Does the paper specify all the training and test details?

Answer: [Yes]



Justification: Reproducibility appendix specifies optimizer/steps for the toy model, the exact  $(r, T, N, B, \text{sweeps, seeds})$  grids per experiment, the reference corpora, and the metric definitions.

## 7. Experiment statistical significance

Question: Does the paper report error bars / statistical significance?

Answer: [Yes]

Justification: The paper’s central claims are the validation-ladder reproductions—exact or near-exact matches to known values (the Domany–Kinzel damage identity, which is verified *bit-exactly*—zero mismatching cells—and its critical points to 0.06–0.15%; logistic-map Lyapunov mean error  $< 10^{-3}$ ; census self-TV 0.22 vs cross 0.95)—which are not null-hypothesis tests, so multiple comparisons do not apply to them. Seed error bars are reported where relevant. Where we *do* test—the ECA class separation and the developmental transition—the unit of analysis is the rule or the run (never the seed-cell), the tests are bootstrap or rank-based, and the developmental family is pre-registered and corrected with Benjamini–Hochberg FDR. The finer three-class ECA ordering was tested, failed ( $p=0.470$ ), and is reported as not surviving. The cross-level conclusion is a *negative* reached by sign-consistency across two model families and two within-model axes, not a surviving  $p$ -value, so no multiplicity correction is load-bearing. The demoted capacity gap is disclosed as a suggestive two-seed effect and not built upon; an optional seed-level rebuild is tracked in the repository (issues #1–#2, #10).

## 8. Experiments compute resources

Question: Does the paper provide sufficient information on compute resources?

Answer: [Yes]

Justification: All experiments run on a single Apple M1 Pro (16 GB); per-phase wall-clock runtimes are tabulated. The toy model uses CPU JAX; real models use fp16 on the MPS backend. Total compute is a few laptop-days.

## 9. Code of ethics

Question: Does the research conform with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: Measurement/interpretability work on public pretrained models and synthetic data; no human subjects, no sensitive data, no new capability that raises misuse concerns.

## 10. Broader impacts

Question: Does the paper discuss potential positive and negative societal impacts?

Answer: [Yes]

Justification: See the Responsible use section. Positive: black-box measurement of trained-LM token dynamics without weight access, applicable to closed/API-only models. It creates no new generative capability. The negative case we treat as most likely is misreading: the paper’s damping length and light cone describe an in-place-resampling construction, and real autoregressive generation absorbs no injected token error, so they must not be cited as evidence about deployed-decoder robustness. That boundary is stated in the body, not an appendix.

## 11. Safeguards

Question: Does the paper describe safeguards for responsible release of high-risk assets?

Answer: [N/A]

Justification: No high-risk models or datasets are released; only measurement code and analysis results on public models.

## 12. Licenses for existing assets

Question: Are existing assets (code, data, models) properly credited and licenses respected?

Answer: [Yes]

Justification: All assets are publicly released and cited: Pythia and BERT checkpoints (Apache-2.0), GPT-2 (MIT), tinyshakespeare (public-domain source text), WikiText-103 (CC BY-SA 3.0), PyTorch (BSD-3-Clause), HuggingFace Transformers and JAX (Apache-2.0). Usage is research-only and within each licence.

## 13. New assets

Question: Are new assets introduced in the paper well documented?

Answer: [Yes]

407 Justification: The instrument (code), the synthetic-Markov calibration corpora, and the toy  
408 checkpoints are documented in the repository README and the Reproducibility appendix.

409 **14. Crowdsourcing and research with human subjects**  
410 Question: N/A.  
411 Answer: [N/A]  
412 Justification: No human subjects or crowdsourcing.

413 **15. Institutional review board (IRB) approvals**  
414 Answer: [N/A]  
415 Justification: No human-subjects research.

416 **16. Declaration of LLM usage**  
417 Question: Does the paper describe LLM usage if it was an important, original, or non-  
418 standard component?  
419 Answer: [Yes]  
420 Justification: Pretrained LLMs are the object of study — they supply the CA update rule —  
421 and this is documented throughout. Separately, and disclosed here per venue policy: an AI  
422 coding assistant was used substantially in implementing the experiments, in analysis script-  
423 ing, and in drafting. All experimental designs, pre-registrations and claims were reviewed  
424 and verified by the authors against committed result files; several errors introduced during  
425 that process were caught by the repository’s own assertions and are recorded as retractions  
426 in the supplementary findings log.